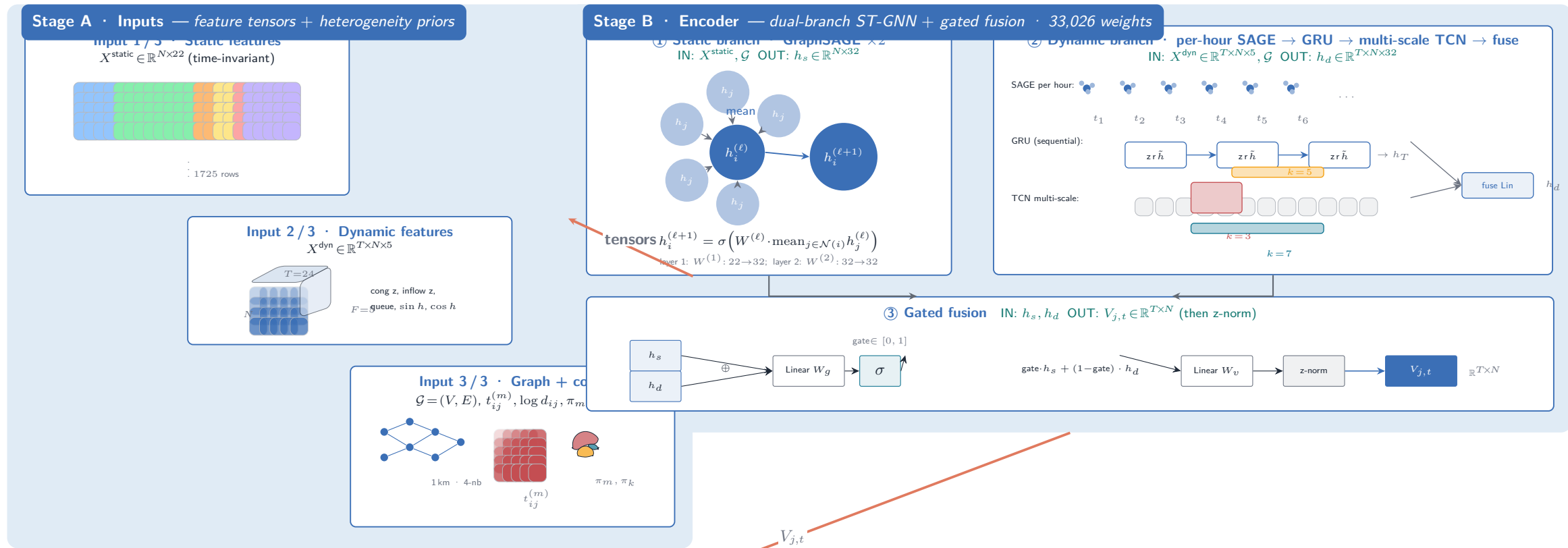


Figure 1 — ML Training Pipeline (*supervised inverse learning of choice parameters from aggregate hourly OD*)



Stage C · Choice head + Loss + Trained Θ — supervised inverse learning · 77 interpretable parameters (0.23%)

④ Choice head — mode $\times$ tier mixture softmax
 IN: $V_{j,t}, t_{ij}^{(m)}, \log d_{ij}, \text{OccM}, \pi_m(i,j), \pi_k(i)$ OUT: $P(j|i, t) \in \mathbb{R}^{T \times N \times N}$

$$U_{ij,t}^{(m,k)} = \alpha V_{j,t} + \beta_{t,m} \kappa_k t_{ij}^{(m)} + \gamma \log d_{ij} + \delta_k \text{OccM}_{ij}$$

$$P(j|i, t) = \sum_{m \in \{\text{car, tr, wk}\}} \pi_m(i, j) \sum_{k \in \{\text{lo, mi, hi}\}} \pi_k(i) \text{softmax}_j U_{ij,t}^{(m,k)}$$

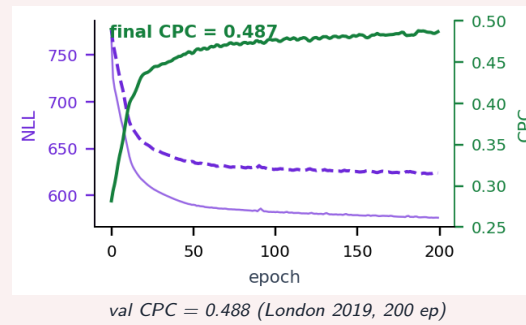
9 softmax cells

⑤ Cross-entropy loss + joint backprop

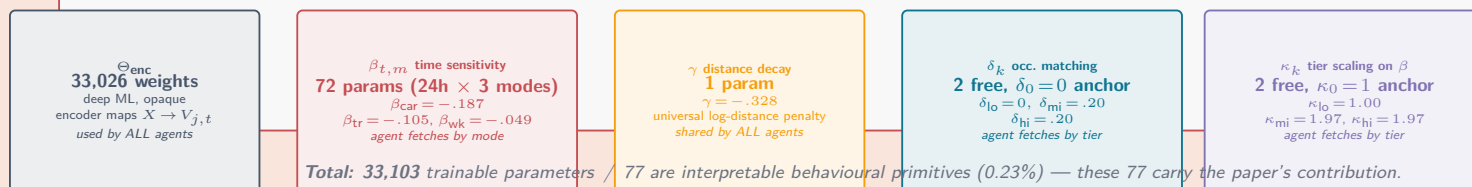
IN: $P(j|i, t), F_{ij,t}^{\text{obs}}$ OUT: trained Θ

$$\mathcal{L} = -\sum_{i,j,t} F_{ij,t}^{\text{obs}} \log P(j|i, t)$$

Adam: enc lr 1e-3, head lr 1e-2; weight-decay 1e-4
 $\nabla_{\Theta} \mathcal{L}$ updates encoder Θ_{enc} AND choice params $(\beta, \gamma, \delta, \kappa)$ jointly



⑥ Trained Θ — output of training, frozen at deployment (bridge to Figure 2)



Trained Θ has 5 functional categories: Θ_{enc} (deep ML, used by anyone going through encoder); $\beta_{t,m}$ (time sensitivity, fetched per agent's mode); γ (distance, shared); δ_k (occ. match, fetched per agent's tier); κ_k (tier scaling on β , fetched per agent's tier). These five parameter types are the bridge to Figure 2 (ABM forward simulation).